

Introduction to Machine Learning course

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Course outline

- Course requirements
 - 3 4 homeworks (theory & programming)
 - Final project
 - Participation
- Rough outline
 - $\frac{1}{3}$ math & probability
 - $\frac{1}{3}$ algorithms & proofs
 - $\frac{1}{3}$ heuristics

Course outline

- What's on the webpage?
- Some parallel courses:
 - Mehryar Mohri (NYU)
 - Avrim Blum (CMU)
- Suggested books – all free!
 - Understanding Machine Learning.
 - Foundations of Machine learning.
 - Neural Networks and Deep Learning
- Data sets
 - Very simple: Hope College
 - Less simple: UCI, MNIST, US government
 - Not simple at all: Kaggle
 - Everything: Google Datasets

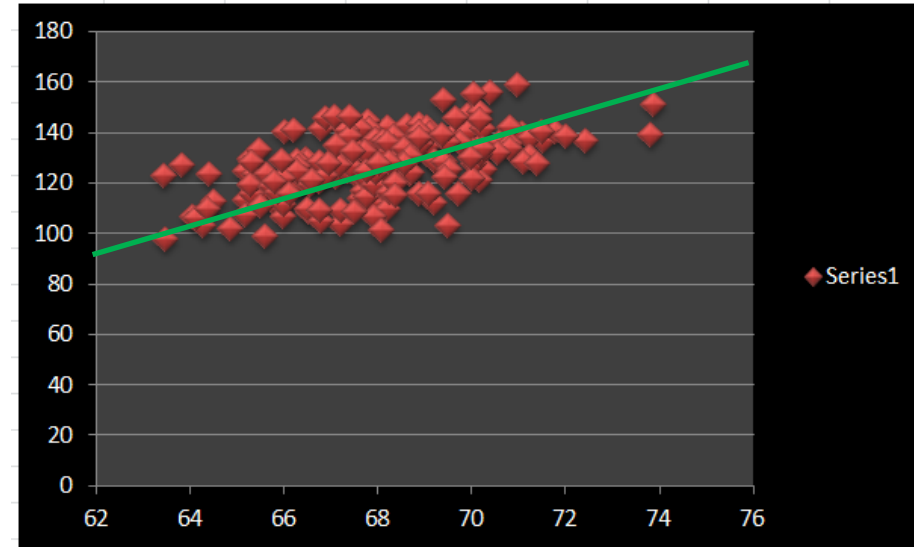
What is machine learning?

- Draw conclusions from data
 - Predict unknown properties
 - Usually from a sample
- For example
 - Given someone's height,
can you guess the weight?
 - Given someone's temperature and heartrate
can you guess the gender?
- Let's look at some Hope College datasets...

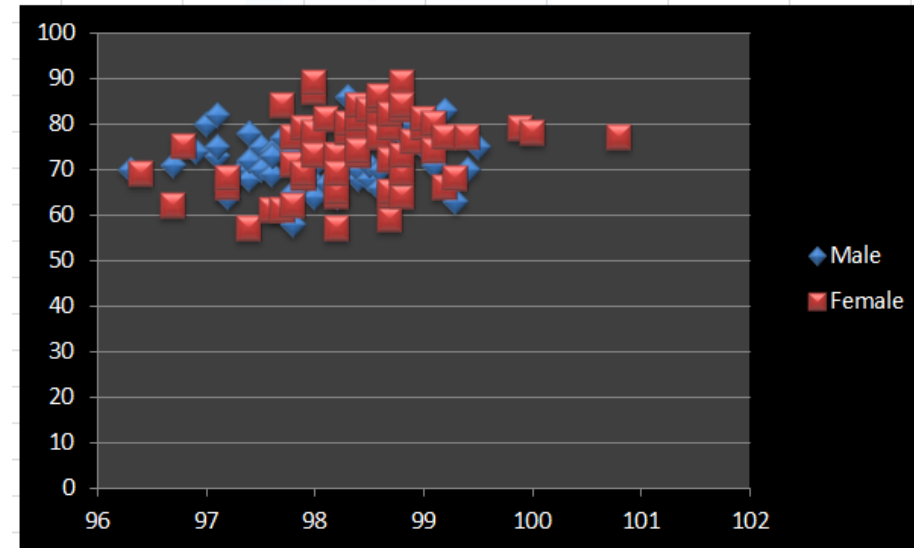


What is machine learning?

- weight vs. height

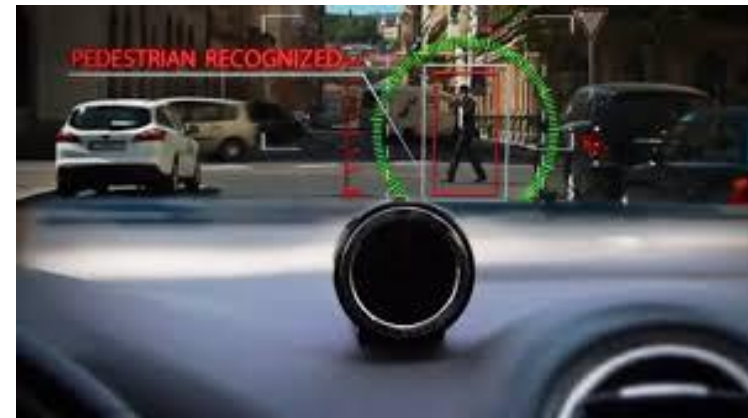


- Temperature & heartrate



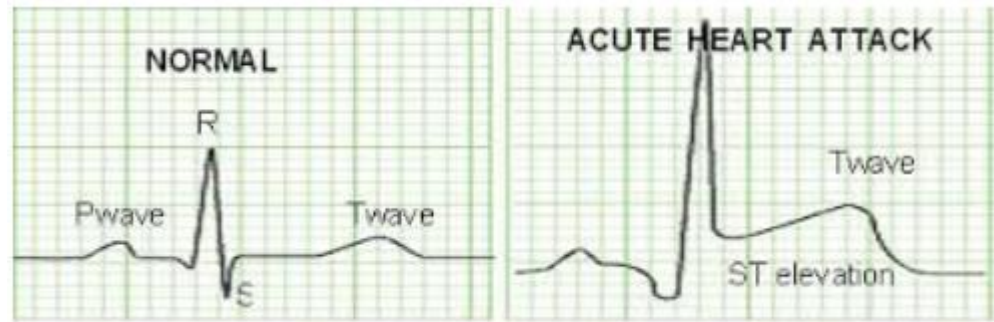
What is machine learning?

- More examples:
- Spam detection
 - Mail server has **datasets** of millions of items that users labelled as spam
 - Can it **extrapolate** to new email?
- Face or image recognition
 - Google has massive **datasets**
 - Can you **extrapolate** to life?

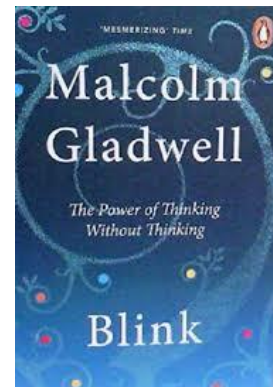


What is machine learning?

- More examples: Automatic EKG reading

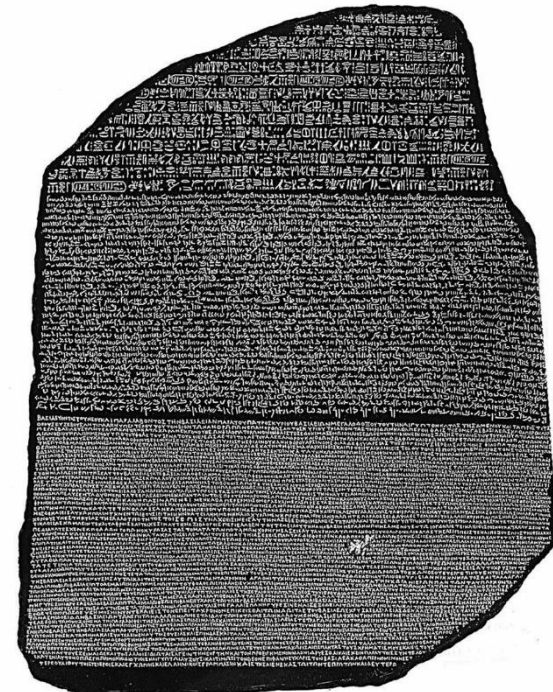


- Consult experts to build model?
 - Are there experts?
 - If there are, do they know the rules?
 - Not robust.
- Goal:
 - Have computer learn it **automatically**
 - using a large labelled dataset



What is machine learning?

- More examples: Automatic translation
- Old approach: **N**atural **L**anguage **P**rocessing
 - Teach computer how language is formed:
 - Verb, noun, adjective...
 - Sentence structure...
- Machine learning goal:
 - Have computer learn it **automatically**
 - using a large dataset of translations
 - Ex: Google translate



What is machine learning?

- Last example: Netflix Prize
 - https://en.wikipedia.org/wiki/Netflix_Prize



What is machine learning?

- Netflix Prize: An example of recommender systems
- Key technique: Principal Component Analysis (PCA)

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5
Comedy	3	1	1	3	1
Action	1	2	4	1	3

	Comedy	Action
User 1		
User 2		
User 3		
User 4		

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5
User 1	3	1	1	3	1
User 2	1	2	4	1	3
User 3	3	1	1	3	1
User 4	4	3	5	4	4

Coin bias

- Let's **learn** something:
 - Given a coin, what's its **bias** p ?
 - If it's a fair coin, its bias is $\frac{1}{2}$
- Each coin toss $x_1, x_2, x_3, \dots$ is a Bernoulli $\{0,1\}$ random variable with
 - $\Pr[1] = p$ $\Pr[0] = 1-p$
- If there are n coin tosses, then we have average value



$$X = (\sum_{i=1}^n x_i)/n$$

Hoeffding's inequality

- Let **B** be a Bernoulli $\{0,1\}$ distribution with parameter p .
- Draw n random independently distributed trials $\sim B$, and let X be their empirical mean
- Then for every $0 < \epsilon < 1$
$$\Pr[|X-p| > \epsilon] < 2e^{-2n\epsilon^2}$$
- We'll prove this (actually Chernoff) next week.
Further reading: Wikipedia

Coin bias

- I tossed a coin **n** times, and saw average

$$X = (\sum_{i=1}^n x_i)/n$$

- Given any $0 < \delta < 1, 0 < \epsilon < 1$, we want to make the following statement:
 - With **probability** at least $1 - \delta$
 - the **bias** of the coin is $X \pm \epsilon$
- How big of a sample do we need?



Coin bias

- By Hoeffding's inequality, probability of failure is at most $2e^{-2n\epsilon^2}$,

- so we need $2e^{-2n\epsilon^2} \leq \delta$

- or $n \geq \frac{\ln\left(\frac{2}{\delta}\right)}{2\epsilon^2}$



- Important: This result is **asymmetric**.
 - Much easier to get better probability (lower δ) than to get tighter bias estimate (lower ϵ)

Coin bias

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– so we need $2e^{-2n\epsilon^2} \leq \delta$

– or $n \geq \frac{\ln\left(\frac{2}{\delta}\right)}{2\epsilon^2}$



$$\begin{aligned} 2e^{-2n\epsilon^2} &\leq \delta \\ -2n\epsilon^2 &\leq \ln(\delta/2) \\ n &\geq -\frac{\ln(\delta/2)}{2\epsilon^2} \\ n &\geq \frac{\ln(2/\delta)}{2\epsilon^2} \end{aligned}$$



- Important: This result is **asymmetric**.
 - Much easier to get better probability (lower δ) than to get tighter bias estimate (lower ϵ)

Memorizer

- Let's do an example that gets to the heart of learning: The Memorizer.
- Suppose you work at a coffee shop
 - You have a week in which you're allowed to ask the patrons whether they want **coffee or tea** (1 or 0)
 - After the first week you need to know what they want without asking
 - Only get paid if you're right
- Abstract learning problem
 - waiter = learner
 - first week customers = sample
 - drink = label
 - payment = measure of success



Memorizer

- What's a good rule?
- Let's try memorizing!
 - For every patron in the first week, write down their name and preference.
 - If someone on the list comes again, give them the same drink.
 - If a new patron comes in, give them a random drink
- Is this a good idea?



Memorizer

- Memorizer success
 - On the sample: 100%
 - On the world: 50%
- Bad idea!



Memorizer

- What's a good rule?
- Suppose you noticed that out of all the hundreds of customers in the first week
 - 95% of men ordered tea
 - 95% of women ordered coffee
- How 'bout this rule:
 - If customer is female, serve coffee. Otherwise serve tea.
- Success:
 - On the sample: 95%
 - On the world: ?



Memorizer

- Most important thing we'll learn in this course:
 - Given a random sample
 - If a **simple** rule is **successful** on the sample
 - Then w.h.p. $(1 - \delta)$ the rule has similar success $(\pm \epsilon)$ on the world.
- **Random sample**: random variables come from same distribution as entire world
- **Simple rule**: Next lectures...
- **Statement**: bias-variance.

Memorizer

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variance



bias

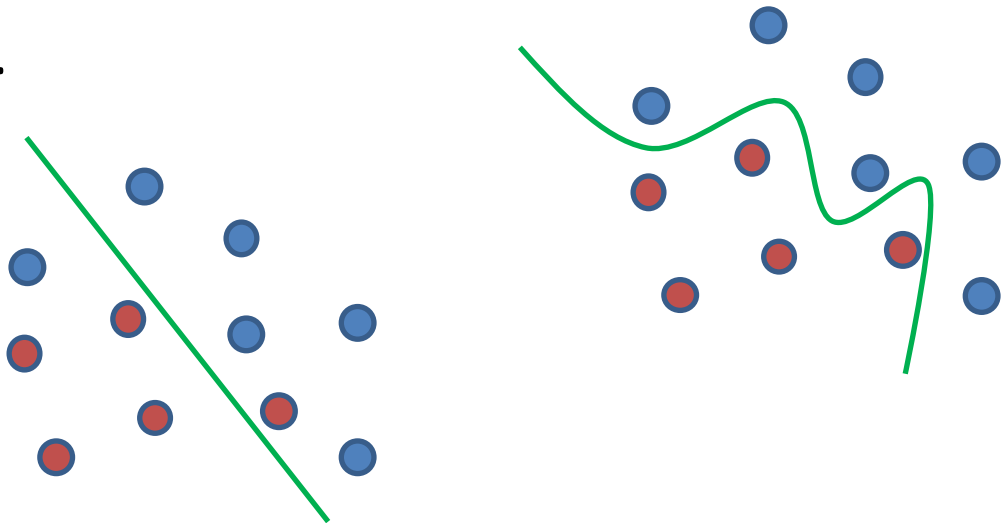
- **Random sample**: random variables come from same distribution as entire world
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Memorizer

- Most important thing we'll learn in this course:
 - Given a random sample
 - If a **simple** rule is **successful** on the sample
 - Then w.h.p. $(1 - \delta)$ the rule has similar success $(\pm \epsilon)$ on the world.
- Can we prove this?
 - Yes. With three classes worth of math and probability.
 - Much time will be spent understanding what is a **simple** rule. How is simplicity measured?

Overfitting

- But what if I find a complex rules that explains the sample well?
 - Like memorizing...



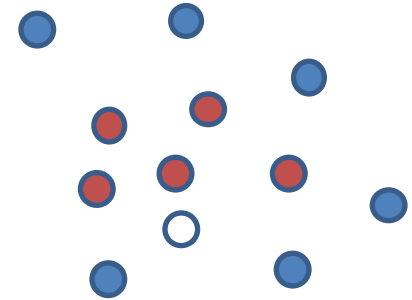
- **Overfitting**
 - Learning the sample, not understanding the underlying rule behind the sample

Learning goals

- Classification
- Multiclass
- Regression
- Ranking
- Clustering
- Dimensionality reduction

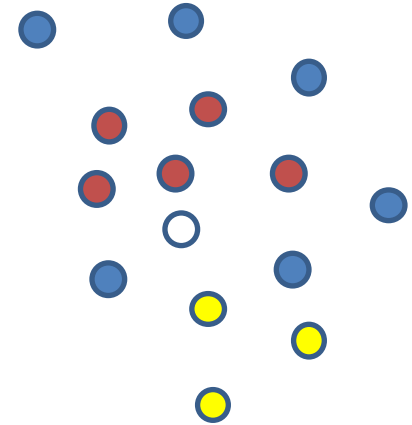
Learning goals

- **Classification** – discrete values
- Multiclass
- Regression
- Ranking
- Clustering
- Dimensionality reduction



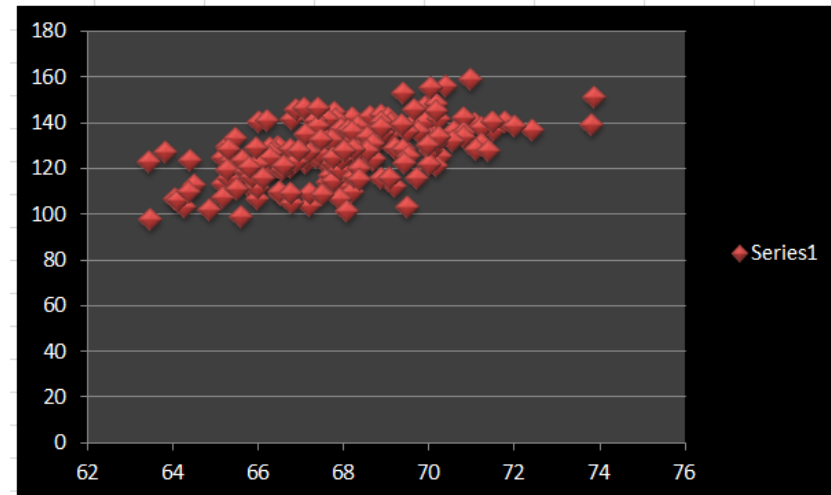
Learning goals

- Classification
- Multiclass – ex: website topics
- Regression
- Ranking
- Clustering
- Dimensionality reduction



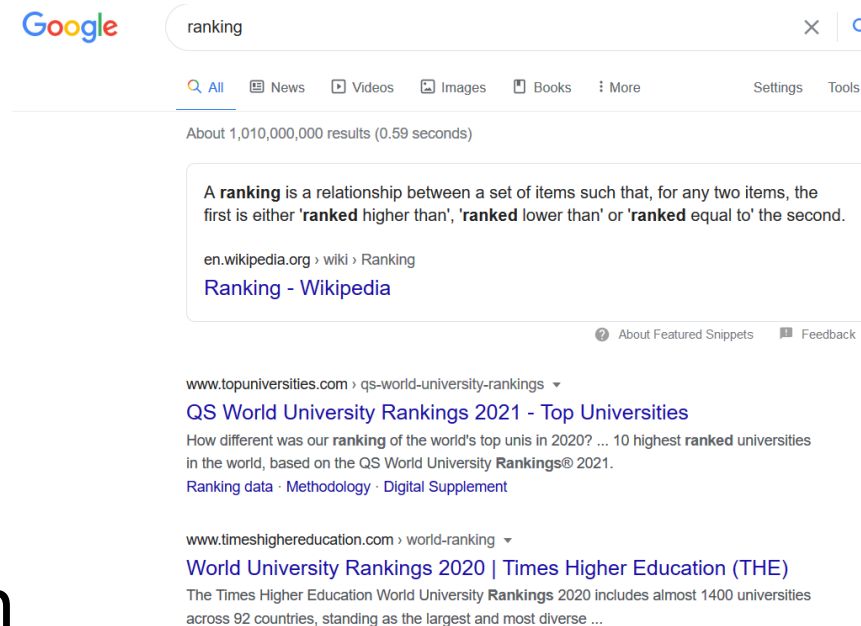
Learning goals

- Classification
- Multiclass
- **Regression** – continuous values: weight/height
- Ranking
- Clustering
- Dimensionality reduction



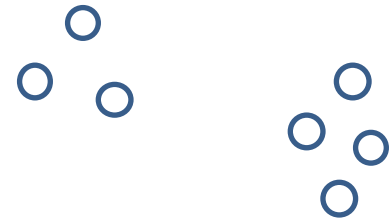
Learning goals

- Classification
- Multiclass
- Regression
- **Ranking – Google search**
- Clustering
- Dimensionality reduction



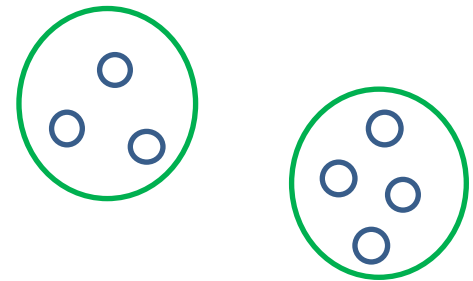
Learning goals

- Classification
- Multiclass
- Regression
- Ranking
- Clustering – unsupervised learning
- Dimensionality reduction



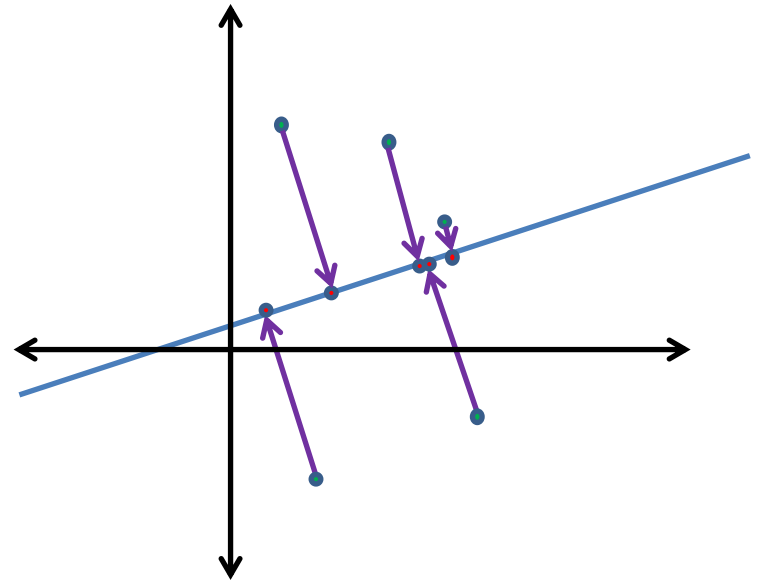
Learning goals

- Classification
- Multiclass
- Regression
- Ranking
- Clustering – unsupervised learning (ex: ads)
- Dimensionality reduction



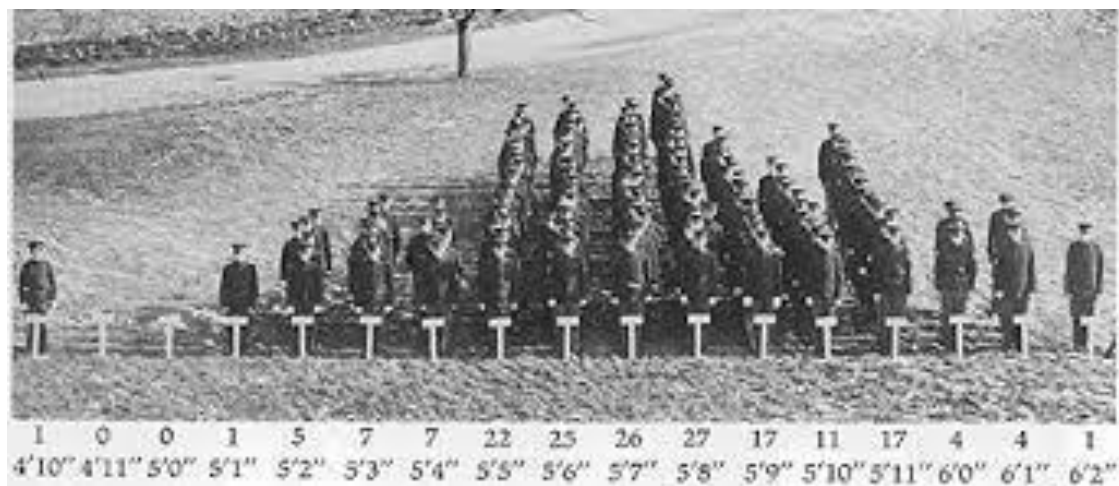
Learning goals

- Classification
- Multiclass
- Regression
- Ranking
- Clustering
- Dimensionality reduction - PCA



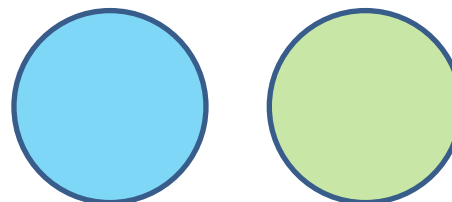
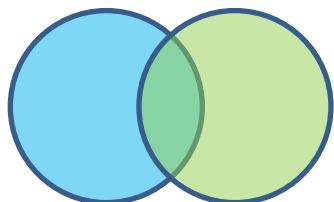
Review of probability

- **Probability distribution**: Description of the probability of every possible event
 - Coins $f(H,T) \rightarrow \{1/2, 1/2\}$
 - Dice $f(1,2,3,4,5,6) \rightarrow \{1/6, 1/6, 1/6, 1/6, 1/6, 1/6\}$
 - Height Normal



Review of probability

- Given two random variable A, B (not necessarily from the same distribution)
 - Conditional: $\Pr[A | B] = \Pr[A \cap B] / \Pr[B]$
 - Bayes' rule: $\Pr[A | B] = \Pr[B | A] \Pr[A] / \Pr[B]$
 - Union bound: $\Pr[A \cap B] \leq \Pr[A] + \Pr[B]$
 - Union bound: $\Pr[A \cup B] \leq \Pr[A] + \Pr[B]$
 - Independent: if $\Pr[A \cap B] = \Pr[A] * \Pr[B]$
or $\Pr[A | B] = \Pr[A]$



Review of probability

Horse	Win %	Gender	color	
Abe	25%	M	White	
Alice	25%	F	White	
Bob	25%	M	White	
Betty	25%	F	Black	

$$\Pr[M | \text{White}] = \Pr[M \cap \text{White}] / \Pr[\text{White}] = \frac{1/2}{3/4} = \frac{2}{3}$$

$$\Pr[M | \text{White}] = \Pr[\text{White} | M] * \Pr[M] / \Pr[\text{White}] = 1 * \frac{1/2}{3/4} = \frac{2}{3}$$

Review of probability

- Given two random variable A,B (not nec. from same dist.)
 - Expectation: $E[A] = \sum_x x \Pr[x]$
 - Dice: $1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3\frac{1}{2}$
 - Linearity: $E[A+B] = E[A] + E[B]$
 - Scale: $E[cA] = cE[A]$
 - Independent: if $E[A \cap B] = E[A] E[B]$
 - Variance: $\text{Var}[A] = E(x - E[x])^2$
 - Scale: $\text{Var}[aA] = a^2 \text{Var}[A]$
- For every random variable X
 - Markov: $\Pr[X \geq tE[X]] \leq 1/t$